

Ernest Katembo Muhasa

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ABOUT ME

Mathematical Sciences and Computer Science graduate with expertise in Machine Learning, Computer Vision, and Deep Learning. Experienced in developing mathematical models, applying physics-informed/data-driven methods to complex problems, and leading tech communities. Seeking to advance research in Artificial Intelligence through a Master's degree focusing on machine learning, Deep learning theory and applied AI.

EDUCATION AND TRAINING

MASTER IN MATHEMATICAL SCIENCES: Math Epidemiology

Africa Institute for Mathematical Sciences (AIMS Cameroon) [31/07/2025 – 01/02/2027]

City: Limbe | **Country:** Cameroon | **Website:** www.aims-cameroon.org | **Field(s) of study:** Natural sciences, mathematics and statistics

Thesis: Physics-Informed Neural Networks for Inverse Problems in Time Dependent Compartmental Models: Parameter Estimation and Forecasting of COVID-19 Dynamics in Canada.

Relevant coursework: Supervised & Unsupervised Machine Learning, Deep Learning Methods, Bayesian Methods, Statistical Inference, Computational Methods for Epidemiology, Infectious Disease Dynamics.

BACHELOR OF SCIENCE (B.S.) IN COMPUTER SCIENCE (5-YEAR UNIVERSITY DEGREE)

Catholic University of Bukavu [30/09/2018 – 30/11/2023]

City: Bukavu | **Country:** Democratic Republic of the Congo | **Website:** www.ucbukavu.ac.cd | **Final grade:** Distinction

Selected courses: Linear Algebra | Numerical Methods | Probability and Statistics | Data Analysis and Machine Learning | Discrete Mathematics | Operational Research | Web programming. | programming R,Python, C++, Java, JS | networks and Telecom

WORK EXPERIENCE

 **MRC Unit The Gambia at LSHTM – Fajara, The Gambia**

Research Assistant Intern (Data Science)

[08/2026 – Current]

 **Catholic University of Bukavu – Bukavu, Democratic Republic of the Congo**

Tutoring Assistant

[31/12/2024 – Current]

- Assisted in teaching computer science courses (Tutoring)
- Supervised lab sessions and guided students on practical projects.
- Developed course materials, graded assignments, and exams.
- Provided academic support and mentorship to students.
- Assisted in organizing research workshops and seminars.

[01/2026 – Current]

Deep Learning Fundamentals Lab

WorldQuant University - New Orleans , Louisiana , United States

- Developed neural network models for diagnosing heart disease using clinical data and supervised learning techniques.
- Built deep learning models for protein localization site classification, applying data preprocessing, neural network training, and multi-class classification.
- Trained neural architectures for image classification, exploring model design, optimization, and performance evaluation.
- Developed a Convolutional Neural Network (CNN) for pet breed classification using image data.
- Applied advanced computer vision models to classify plant seedlings, gaining experience in image preprocessing, model training, and evaluation.
- Implemented Recurrent Neural Network (RNN)-based models for human activity recognition using sequential and time-series data.
- Implemented autoencoders and variational models for representation learning and generative modeling.
- Developed Generative Adversarial Networks (GANs) for artificial image and artwork generation.
- Strengthened practical skills in Python, PyTorch, deep learning architectures, model optimization, and performance evaluation through hands-on projects.

Write here the description...

[03/2025 – 06/2025]

Applied AI Lab: Deep Learning for Computer Vision

WorldQuant University - New Orleans , Louisiana , United States

- Worked on real-world computer vision projects using deep learning.
- Developed and trained CNNs, object detection, and image classification models.
- Applied techniques like transfer learning, data augmentation, and model optimization.
- Used tools such as TensorFlow, Keras, OpenCV, and PyTorch.
- Gained hands-on experience in image processing, feature extraction, and model evaluation.

[06/2024 – 03/2025]

Applied Data Science Lab

WorldQuant University - New Orleans , Louisiana , United States

- Completed eight real-world data science projects, applying machine learning, data analysis, and predictive modeling.
- Worked with real datasets to solve business and social challenges.
- Gained hands-on experience in data cleaning, feature engineering, model building, and evaluation.
- Used tools like Python, SQL, Pandas, Scikit-learn for data-driven solutions.
- Developed strong problem-solving skills through practical case studies.

CONFERENCE PRESENTATION

[17/08/2025 – 22/08/2025]

Deep Learning Indaba 2025

[04/02/2025 – 06/02/2025]

Global Students Research Conference

King Fahd University of Petroleum and Minerals – Dhahran, Saudi Arabia

Abstract: This study develops a mathematical model to improve waste management in Bukavu. It optimizes collection routes, predicts waste generation, and suggests cost-effective recycling and disposal strategies. Using data-driven methods, the model helps policymakers create sustainable solutions for the city's waste challenges.

[04/2023]

Institute for Advanced Teaching and Learning (IATL) - WorldCUR & BCUR 2023

Warwick University – London, United Kingdom

Abstract: In light of the COVID-19 pandemic, effective hand hygiene has become a critical measure to reduce virus transmission. This study presents the design and simulation of an automatic handwashing system implemented at the General Provincial Reference Hospital of Bukavu, leveraging Arduino technology to facilitate touch-free hand hygiene. The system comprises an ultrasonic sensor, Arduino microcontroller, relay module, and submersible water pump to ensure that users can wash their hands without physical contact, significantly reducing the risk of hand-transmitted infections. By addressing hygiene challenges, particularly in healthcare settings with limited resources, this innovation contributes to enhanced infection control and demonstrates a viable solution for preventing the spread of COVID-19 and other communicable diseases.

Link: <https://bit.ly/3Rslvdx>

LEADERSHIP, MENTORING AND SERVICE

[13/09/2021 – Current]

Google Developer Groups Kivu (Bukavu & Uvira) - Organizer

Analytical thinking | Community management | Content creation | Digital literacy Education | Leadership

Links: <https://gdg.community.dev/gdg-kivu/> | <https://gdg.community.dev/gdg-uvira/>

[09/02/2022 – Current]

Veilleurs du Web RDC - Secretary/ Bukavu

In Charge of Information and Communication Technologies

Community Management | Team Collaboration | Public Speaking | Advocacy and Innovation

[04/2022 – 05/2023]

Student Representative - Catholic University of Bukavu

Catholic University of Bukavu

Development | Language training

Link: <https://bit.ly/45rXSY1>

[05/2020 – 10/2023]

President - English Club of Catholic University of Bukavu

Google Developer Student Clubs Catholic University of Bukavu

[07/2022 – 08/2023]

Co-Leader

Google Developer Student Clubs Catholic University of Bukavu

Link: <https://bit.ly/4bcaSTe>

[06/2021 – 07/2022]

Leader

Link: <https://bit.ly/3Xo7LnS>

SELECTED HONOURS AND AWARDS

[08/2025 – 02/2027]

MasterCard Scholarship at AIMS Cameroon

[11/2023]

Certificate of Merit of Catholic University of Bukavu

Project: Prediction of diseases using Machine Learning.

Data Analysis | Machine learning | Artificial Intelligence | Python programming | Django Framework

[03/2021 – 04/2023]

Google Career Certificate Scholarship (GCCS)

Online learning on Coursera Platform for the course of IT Support and IT Automate

[30/09/2021 – 31/10/2022]

Excellence Scholarship of Catholic University of Bukavu

- Mastercard Scholarship at the African Institute for Mathematical Sciences
- The scholarship awarded for achieving 80% on an academic Year

SELECTED SKILLS

Soft skills

Research and analytical skills, Written and Verbal Communication, Organizational and planning skills, Presenting, etc..

Language skills

English(Advanced), French (Advanced), and Lingala, Swahili (Native)

Digital and Software

- Programming languages: R, Python, JS
- Data analysis and visualization, Machine learning / AI (ML, Deep Learning, NPL, CV, ...)
- Microsoft Excel, Word, Powerpoint, Latex

I certify that the information provided is accurate and true to the best of my knowledge. References and supporting documents are available upon request.

Banjul, 24/08/2026



Ernest Katembo

Muhasa